

Question	Answer	Mark	Mark scheme	Additional guidance
20	$y + \sqrt{3}x = 4$	P1 P1 P1 A1	for process to find the value of p, eg $\sqrt{4-1^2} (= \sqrt{3})$ for a start of a process to find gradient of tangent, eg gradient of normal/radius = $\frac{1}{p}$ or $\frac{1}{\sqrt{3}}$ or $\frac{1}{[p]}$ or for gradient of tangent = $-p$ or $-\sqrt{3}$ or $-[p]$ (dep P1) for substituting ("$\sqrt{3}$", 1) into $y = -\sqrt{3}x + c$ or for $y-1 = -\sqrt{3}(x-\sqrt{3})$ oe or for $1 = -p \times p + c$ or for substituting ($[p]$, 1) into $y = -[p]x + c$ or for substituting ("$\sqrt{3}$", 1) into $y = -\frac{1}{[m]}x + c$ for $y + \sqrt{3}x = 4$	May occur later in the process Where $[p]$ is their stated value of p Where $[m]$ is clearly their gradient of the normal/radius A correct answer with no supportive working gets 0 marks